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Local Failure After Episcleral Brachytherapy for Posterior Uveal Melanoma: Patterns, Risk Factors, and Management



EDITOR:

HAVING READ THE PAPER “LOCAL FAILURE AFTER EPISCLERAL brachytherapy for posterior uveal melanoma: Patterns, risk factors, and management,” we are compelled to clarify the science, for those interested.¹

Clearly, recurrences following brachytherapy need to be reported. However, we do not agree that rates of recurrence are the same for all forms of plaque brachytherapy. Unlike our 2016 publication that included normative treatments used at 10 worldwide subspecialty centers, this paper combines 2 forms of plaque brachytherapy and methods unique to the Cleveland Clinic.²

Regarding different types of plaques, high energy beta rays from ¹⁰⁶Ru plaques exhibit unique dose distributions and hence distinctive effects as compared with the low-energy photons emitted from ¹²⁵I. For example, beta particles travel relatively short distances in tissue, leading to very steep dose gradients at depth and relatively sharp peripheral margins.^{3,4} Experienced ¹⁰⁶Ru surgeons compensate by employing larger

tumor-free margins (larger plaques) and higher apical prescription doses.⁵ Failure to do so has led to high local recurrence rates.⁵

Regarding case selection, uveal melanoma case selection is important. As indicated by the 2014 consensus guidelines of the American Brachytherapy Society (ABS) and the American Association of Physicists in Medicine (AAPM), tumors greater than 6 mm in height risk failure when treated with the beta-irradiation source ¹⁰⁶Ru.^{3,4} In contrast, much taller tumors could be irradiated with ¹²⁵I and ¹⁰³Pd plaques.^{3,4}

The authors report that patients “underwent brachytherapy with ¹⁰⁶Ru or ¹²⁵I plaque for posterior uveal melanoma (ciliary body or choroidal).” Ciliary body melanomas are not posterior, and surgical placement of anterior plaques is easier and less likely associated with mis-placement. In this series, the predominant site of recurrence was at the margins, suggesting the major risk factor for recurrence was suboptimal plaque placement.³

In this paper, the authors failed to describe their tumor size as related to plaque selection (¹²⁵I or ¹⁰⁶Ru) or their radionuclide-specific radiation dose. In addition, Table 4 shows that over 90% (340/375) received either primary or secondary transpupillary thermotherapy.⁴ Laser-treated tumors should have been either excluded from the study or factored into the title and conclusions.

Regarding the literature, no author wishes to be misquoted in the AJO. In Table 1, the authors misquote Dr Finger’s 2009 paper on 400 cases of ¹⁰³Pd plaque therapy outcomes.⁶ The authors claim that the 3% of our local control failures were treated by enucleation. In fact, only 1.2% (5/400) required or requested eye removal for recurrent disease. The other 7 had eye- and vision-sparing treatment.

In order to compare this work to others, the authors should use the latest American Joint Committee on Cancer/Union International for Cancer Control staging systems.⁷ This paper uses the 35-year-old Collaborative Ocular Melanoma Study guidelines.

In conclusion, it does not serve the ophthalmic literature to treat all forms of irradiation as equivalent. Therefore, we suggest that ophthalmic journals be held to the same standards as radiation therapy journals in reporting the results of ophthalmic radiation therapy. Specifically, both should require more detailed description of radiation source selection and methods of application in comparison to consensus standards.

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FUNDING/SUPPORT: RESEARCH WAS SUPPORTED BY THE EYE Cancer Foundation, Inc (<http://eyecancercure.com>) (P.T.F.). Financial Disclosures: The following authors have no financial disclosures: Paul T. Finger, and Wolfgang A.G. Sauerwein. The authors attest that they meet the current ICMJE criteria for authorship.